



Stochastic Modeling of Glaciar Melting via Latin Hypercube Sampling (LHS)

Course: CSE400 – Fundamentals of Probability in Computing

Domain: Climate

Milestone 4

GROUP-12

Dhairya Shah, Dhanvi Joshi, Jahnavi Patel, Jashvi Shah, Manya Morakhia





Objective and Importance

Instead of drawing completely independent random samples as in standard Monte Carlo, implementing Latin Hypercube Sampling (LHS) to generate the climate input variables.

Climate Variables are stochastic. Hence their variability needs to be considered . Thus random sampling model like Monte Carlo Simulation and LHS are used.

Monte Carlo Simulation Overview

Milestone 3

Implemented a purely random Monte Carlo simulation with independent draws for all climate variables.

Key Limitations:

- Data Clumping: Random draws often cluster in the center of the distribution, leaving critical tail risks unexplored.
- Inefficiency: Requires extremely large sample sizes to achieve a stable mean.
- Variance: High fluctuation between runs makes the Risk Score less reliable for planning.

System Pipeline Overview

Climate Distributions



LHS Stratification Matrix



3-Stage Physics Engine



Probabilistic Results.

The core decision rule operates at the LHS Transformation stage, where N samples are paired to form unique input vectors.



Sources of Uncertainty

There are mainly 2 Sources of uncertainty

- Variables: Temperature, Precipitation, Albedo, Cloudiness, and Snow Threshold.
- Monte Carlo Simulation: Standard Monte Carlo Simulation may entirely miss the Perfect Storm scenarios which means maximum heat paired with minimum albedo due to the randomness of sampling. Due to this uncertainty we moved from Monte Carlo Simulation to LHS.



Key Random Variables

Five environmental variables were modeled probabilistically:

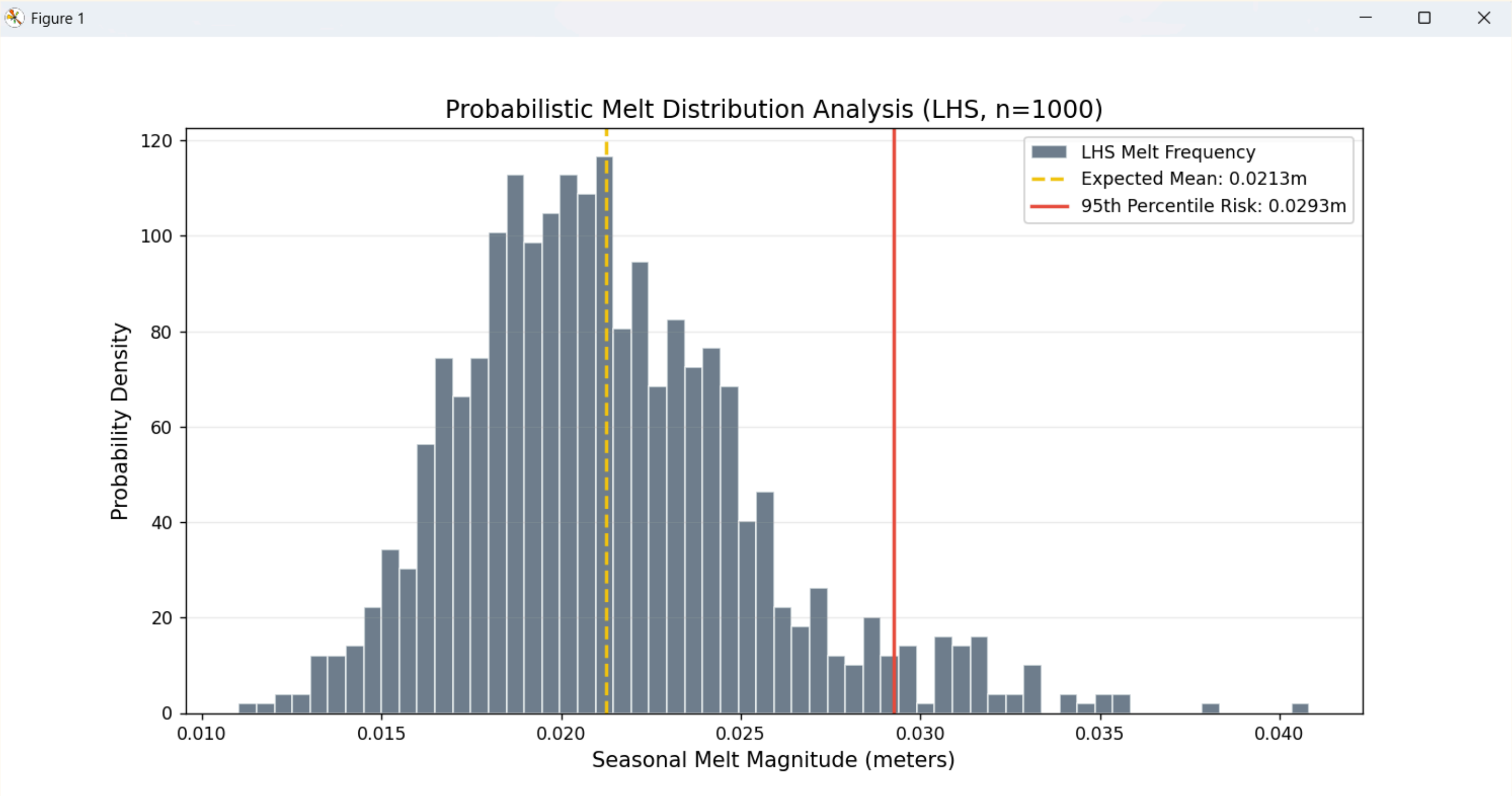
Random Variables	Distribution Type	Role in Simulation
Precipitation	Gamma	Influences snow accumulation and melt dynamics
Ice albedo	Truncated Normal	Controls solar radiation absorbtion
Snowfall threshold	Normal	Influences glacier mass balance
Cloudiness	Normal	influences surface energy balance and total glacier melt
Air temperature	Normal	Affects melting energy



Randomized Algorithm Design (LHS Engine)

- The Algorithm: Latin Hypercube Sampling (LHS).
- Mechanism: Each variable's range is divided into equal probability intervals where exactly one sample is drawn from each interval.
- Random Pairing: These samples are then shuffled and paired across 5 dimensions to form a 5000 X 5 input matrix.

Main LHS Graph





Probabilistic Reasoning Behind the Algorithm

- **Core Idea:** LHS reduces randomness-induced variance by enforcing structured sampling.
- **Variance Theory:**
 - Monte Carlo convergence: $O(1/\sqrt{N})$
 - LHS convergence: $O(1/N)$
- **Why Better:**
 - Ensures full distribution coverage
 - Captures extreme (“tail”) scenarios systematically
- More stable estimation of melt without increasing sample size

Implementation of the Randomized Strategy

- **Step 1:** Generate 5000x5 LHS matrix in [0,1]
- **Step 2:** Transform using Inverse CDF (PPF):
 - Normal $\rightarrow$ norm.ppf()
 - Gamma $\rightarrow$ gamma.ppf()
- **Step 3:** Form input vectors: [Ta,P, α ,n,Ts]
- **Step 4:** Pass into 3-stage physics engine

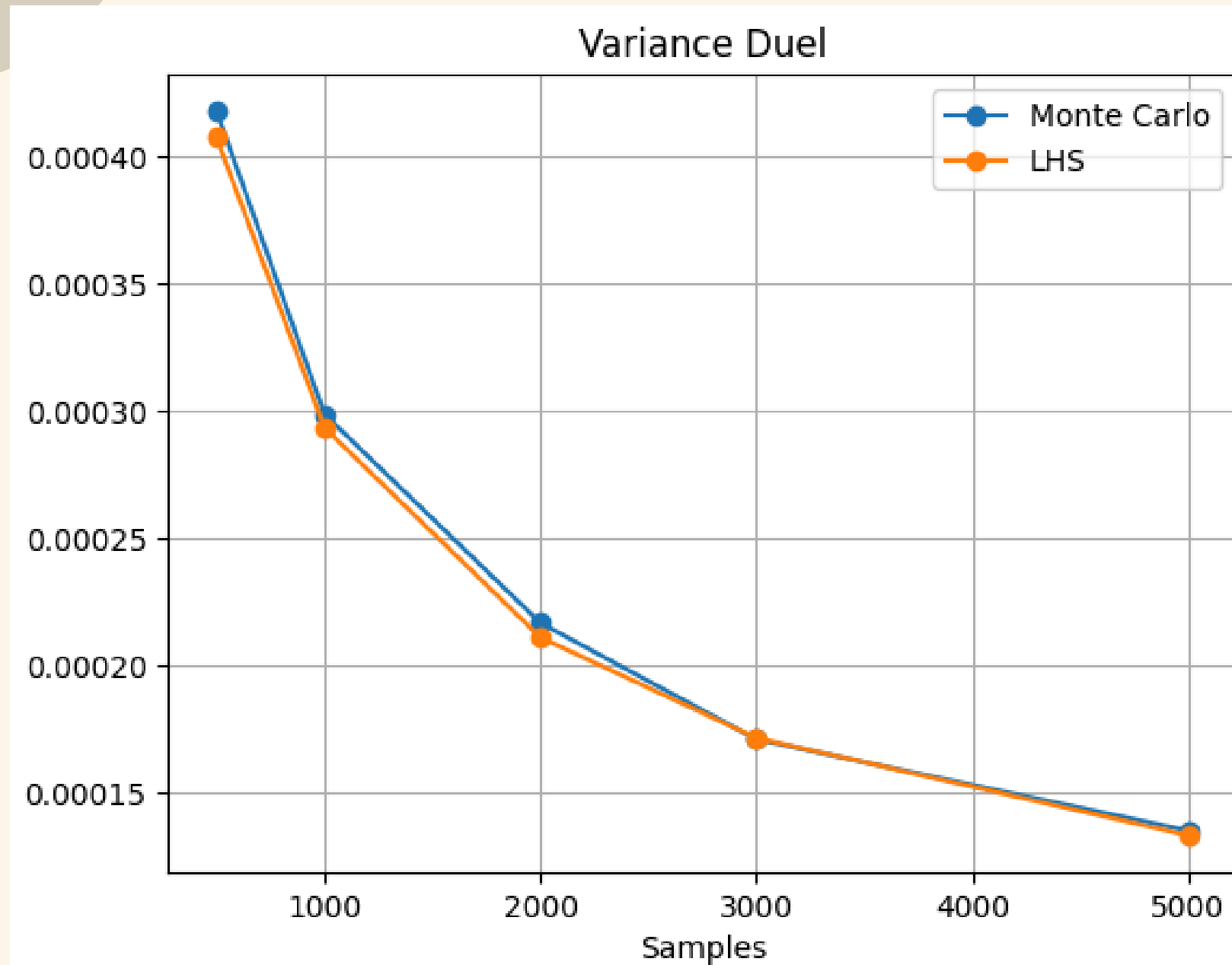
Ta1	P1	α1	n1	Ts1
Ta2	P2	α2	n2	Ts2
⋮	⋮	⋮	⋮	⋮
Tak	Pk	αk	nk	Tsk



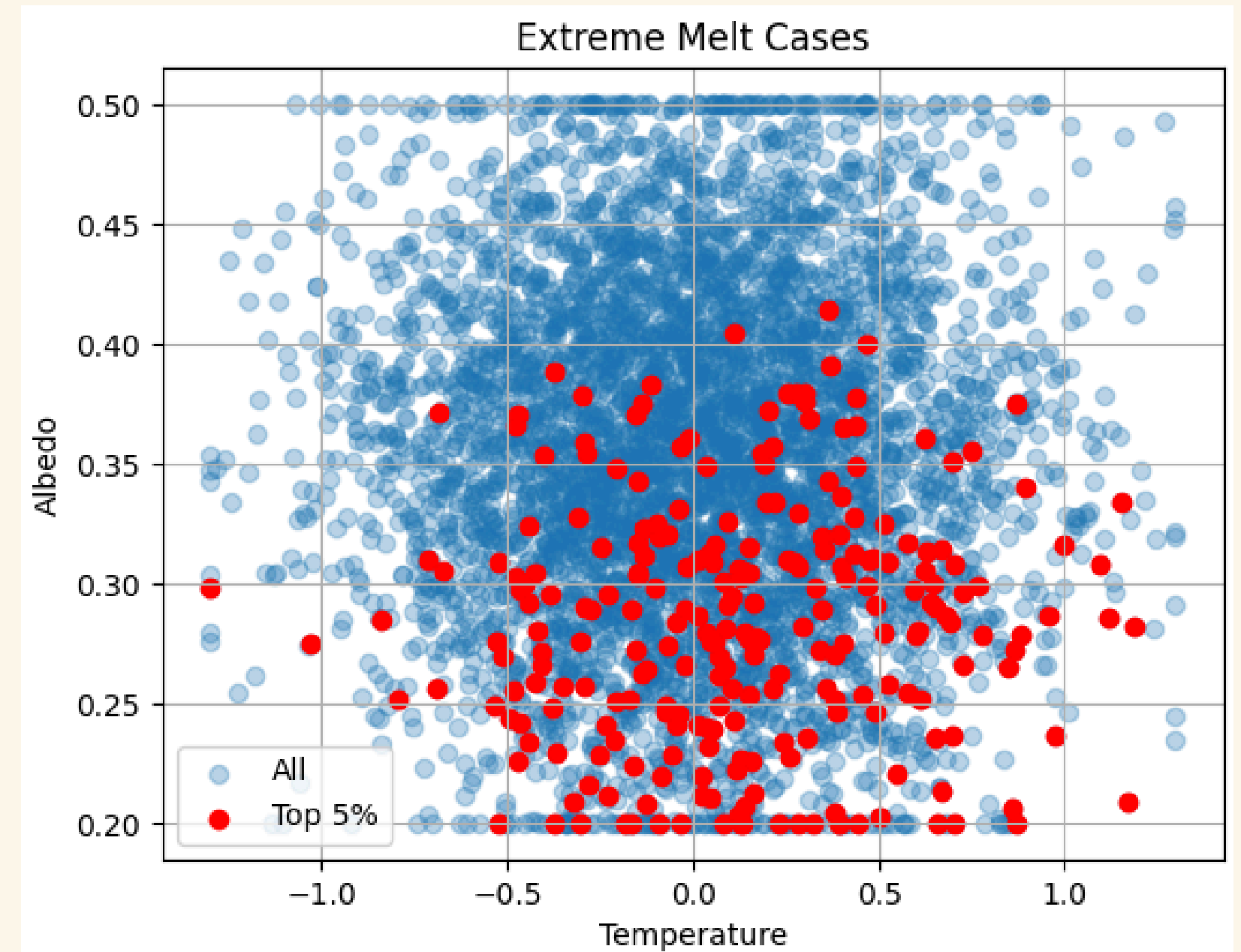
Evaluation Metrics

- **Mean Melt Estimate** → Overall system output
- **Variance of Output** → Stability indicator
- **Standard Error of Mean:**
 - $\sigma_{\bar{X}} = \sigma / \sqrt{N}$
- **95th Percentile (Risk Score):**
 - Defines extreme melt scenario
- **Why These Metrics:**
 - Measure both **accuracy (mean)** and **reliability (variance)**

Comparative Results (MC vs LHS)



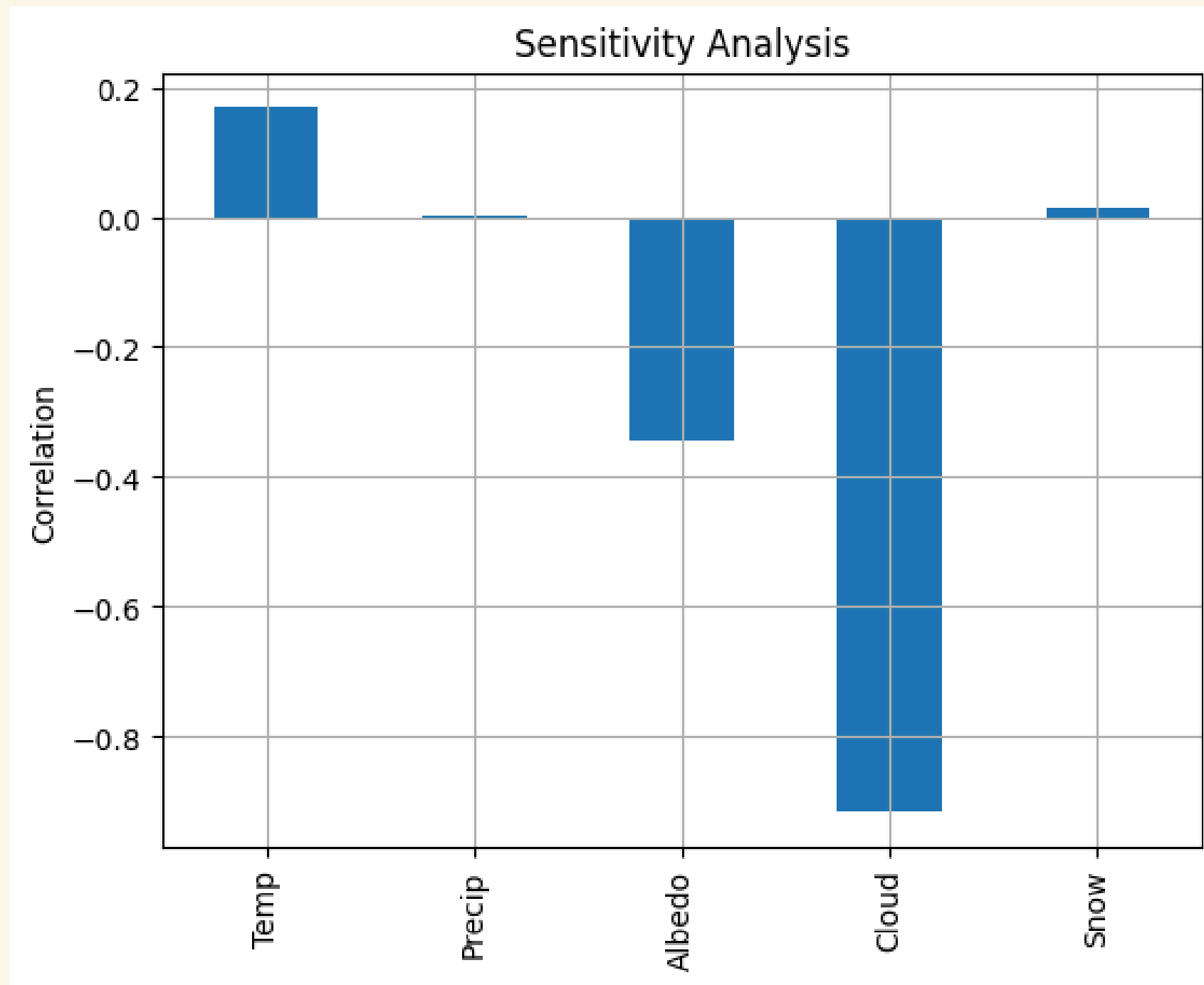
LHS shows lower standard error than Monte Carlo



Identifies high-risk melt scenarios

Sensitivity Analysis

This graph shows that Air Temperature and Ice Albedo contribute the most to melt variability, making them the primary drivers of extreme risk.





Key Insights

- LHS ensures systematic exploration of all probability intervals
- Variance Duel: LHS produces more stable estimates than MC
- Risk Interpretation:
 - 95th percentile defines extreme melt scenario
- Final Takeaway:
 - LHS improves both efficiency and reliability of simulation



Thank You

